Example 6.13 The quantity demanded of a certain type of Compact Disk is 22,000 units when a unit price is ₹ 8. The customer will not buy the disk, at a unit price of ₹ 30 or higher. On the other side the manufacturer will not market any disk if the price is ₹ 6 or lower. However, if the price ₹ 14 the manufacturer can supply 24,000 units. Assume that the quantity demanded and quantity supplied are linearly proportional to the price. Find (i) the demand equation (ii) supply equation (iii) the market equilibrium quantity and price. (iv) The quantity of demand and supply when the price is ₹ 10.

Solution:

Let the x-axis represent the number of units in thousands and the y-axis represent the price in rupees per unit.

(i) For demand function, let (x_1, y_1) and (x_2, y_2) be (22,8) and (0,30) respectively.
Using two point form, we get the demand function as

$$\frac{y-8}{30-8} = \frac{x-22}{0-22}$$
$$\Rightarrow y_D = -x + 30 \text{ (demand function).}$$

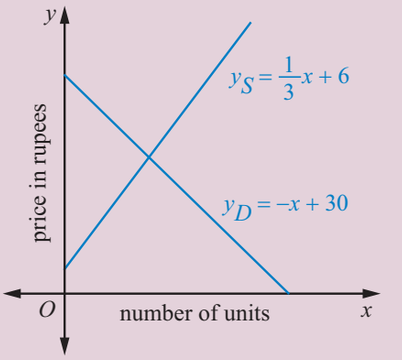


Figure 6.29

(ii) For supply function, let (x_1, y_1) and (x_2, y_2) be (0,6) and (24,14) respectively.
Using two point form, we get the supply function as

$$\frac{y-6}{14-6} = \frac{x-0}{24-0}$$
$$y_S = \frac{1}{3}x + 6 \text{(supply function).}$$

(iii) At the market equilibrium the demand equals to supply,
That is, $y_D = y_S \quad \Rightarrow -x + 30 = \frac{1}{3}x + 6$

$$x = 18 \text{ and } y = 12.$$

Market equilibrium price is Rs12 and number of quantity is 18,000 units.

(iv) when the price $y = 10$, from the demand function $y_D = -x + 30$, we get $x = 20$.
That is, the demand is 20,000 units.
Similarly from the supply function $y_S = \frac{1}{3}x + 6$, we get $x = 12$.
Hence, the supply is 12,000 units.

Example 6.14 Find the equation of the straight line passing through $(-1, 1)$ and cutting off equal intercepts, but opposite in signs with the two coordinate axes.